Regular Prices vs. Discounts: How Pricing Strategies Perform Across Store Formats *

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Abstract

This study investigates how regular prices listed and discounts offered (i.e., temporary price reduction) influence brand sales across store formats within a single grocery retail chain. Using loyalty-based scanner data covering 55 brands across 14 categories from 2014–2016, we estimate error-correction models to quantify elasticities of regular prices and discounts in a hypermarket, a supermarket, and a convenience store. Regular price elasticities are negative across all store formats, with the strongest magnitude in the hypermarket and the weakest in the convenience store. Discount elasticities are positive and highest in the hypermarket, followed by the supermarket, indicating stronger promotional responsiveness in larger-format stores. The differential elasticities show that the impact of regular price increases generally exceeds that of discounts, suggesting asymmetry in customer responses. The second-stage results show that store formats significantly influence how brand and category factors determine pricing effectiveness. Results highlight that pricing effectiveness varies systematically by store format and that separating regular price and discount effects provides a better understanding of promotional pricing across retail environments.

Keywords: Discounts; Pricing; Grocery retailing; Store formats

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1 Introduction

Retailers in the grocery industry constantly face changing consumer preferences and shopping behaviors driven by dynamic micro and macroeconomic conditions (Scholdra et al., 2022). These conditions significantly affect households' shopping behavior, leading to adjustments in where and how much consumers shop and spend. In response to these shifts, many adopt multiformat retailing, where different store formats, such as hypermarket, supermarket, and convenience store, are designed to meet diverse needs, broaden reach, and enhance the shopping experience (Breugelmans et al., 2023; Dekimpe et al., 2023). Each store format facilitates distinct shopping goals, thus attracting customers with varied behaviors and preferences (gijsbrechts et al., 2018; Jindal et al., 2020).

Given these differences in consumer behavior across store formats, the effect of marketing mix instruments, particularly price and discount, are likely to vary (e.g. Shankar & Krishnamurthi, 1996; Haans & Gijsbrechts, 2011; Widdecke et al., 2022). While many studies have explored price and discount effects across brands, categories, or store formats (e.g., Pauwels et al., 2007, ;Haans & Gijsbrechts, 2011; Datta et al., 2022), few have considered all these factors systematically together. Examining price and discount effects simultaneously across brands, categories, and formats provides a more complete view of how these elements interact to influence brand performance, particularly for multiformat retailers who have to manage and navigate the complexities of multiple consumer environments.

Promotional pricing, a retail strategy involves either adjustments of the regular (baseline) price or the offering of temporary discounts, has been extensively studied, though the approaches to its analysis vary. Research typically integrates these adjustments into a 'final price' or treats them separately to address the distinct cognitive processing involved in evaluating regular prices and discounts (Kuntner & Teichert, 2016). Both regular price and discounts are important in consumer decision-making, but when analyzed together, distinct consumer responses to each may be overlooked. Neglecting the differences between them may also lead to biased estimates regarding price sensitivities (Lu et al., 2023). Thus, an analysis of how regular prices and discounts distinctly influence consumer behavior across different store formats can support effective promotional planning across various retail environments. For instance, a discount may drive sales for certain brands in supermarkets but have limited effect in convenience stores, reflecting the differential impact of pricing strategies across store formats. In practice, pricing decisions often involve selecting whether to prioritize regular prices or discounts, with choices determined by store format, product category, and brand factors.

The purpose of this study is to investigate the differentiated impacts of regular prices and discounts on brand performance across various categories and store formats within the same retail chain. By distinguishing and quantifying the effects of promoted discounts from the final retail price on brand sales, we contribute to addressing existing gaps in understanding the differential impacts of these components. Specifically, we examine the direct influences of store formats on pricing effectiveness. Pricing effectiveness may also depend on brand- and category-level characteristics, including discount frequency, discount depth, private label status, impulsiveness, storability, and competitive intensity. Therefore, we further analyze how store format moderates the relationships between these factors and pricing effectiveness. To achieve this, we utilize detailed scanner data from 2014 to 2016, obtained through the loyalty membership database of a Nordic grocery retailer operating as a system with individual store owners under the same retail-chain. In this system, promotional strategies are largely centralized, while individual stores have the flexibility to set their own prices.

In empirical analysis, we select 180 brands across 14 frequently purchased categories across store formats engaging in adjustments of the regular prices or the offering of temporary discounts. We follow a two-stage approach. In the first stage, we quantify the elasticities of regular prices and discounts on brand sales across categories and store formats. To mitigate potential bias stemming from differences in shopping intent across formats, we incorporate trip types as an additional control variable. This trip type is classified algorithmically from transactional basket data. Incorporating this construct helps ensure that observed differences in price responsiveness are not confounded by the underlying purpose of the shopping trip. In the second stage, we explore how store formats influence these elasticities as well as moderate how brand and category factors affect these pricing elasticities. Our findings suggest variations in how promotional strategies (i.e., changes in regular prices and changes in discounts) perform across different formats and how effects from brand and category factors are moderated. For example, the significant influence of store formats on the differential effects between regular price changes and discounts, particularly in convenience stores and supermarkets, indicates a unique consumer sensitivity to pricing strategies that is not uniformly mirrored across other store types. Additionally, the role of store formats in moderating brand and category factors, such as discount depth and breadth, especially in hypermarkets and supermarkets, shows that larger store formats can leverage scale and assortment diversity to make promotions more compelling and effective.

The findings of our study not only contribute to the understanding of how different store formats influence the effectiveness of promotional pricing strategies within a multi-format retail setting, but also offer actionable insights for retail managers. For example, the heightened responsiveness to discounts in hypermarkets and supermarkets can be used strategically to increase basket size and improve turnover, particularly for categories known for having wider discounts offered. The distinct effects observed in hypermarkets for private labels, where discounts sensitivity is higher, address the strategic importance of pricing these products correctly. The lack of significant effects from store formats on discount elasticity suggests a uniformity in how discounts are perceived across formats. This finding calls for a more tailored approach to discounting or decentralizing how the discount should be offered.

The rest of the paper is organized as follows. We provide a brief overview of relevant literature and our research framework in Section 2. We introduce the empirical setting, data and measure for this study in section 3 and present model-free evidence in section 4. We specify our two-stage approach to model in section 5. We present and discuss our results in Section 6 and 7 respectively.

2 Background and Research Framework

2.1 Literature Review

Previous studies have explored the effects of store formats, brand, and category factors on promotional pricing effectiveness. However, a comprehensive analysis that fully investigates the potential role of store formats in directly influencing promotional price effectiveness and moderating the impact of brand and category factors is limited. Table 1 provides an overview of the relevant literature on promotional pricing effectiveness, highlighting the key factors that influence this effectiveness and the various approaches to analyzing the differential effects of regular prices and discounts.

Not all research, however, has specifically focused on the role of physical store formats within the same retail chain in influencing the elasticities of promotional prices, nor have they extensively examined how brand and category factors can be affected differently across formats. Since promotional pricing typically involves either adjustments to the regular (baseline) price or the application of temporary discounts aimed at creating a perceived value for the consumer (van Heerde & Neslin, 2017), academic research has examined promotional pricing in different aspects—either by analyzing regular prices and discounts together (e.g., Pauwels et al., 2007; Datta et al., 2022) or by treating them as separate elements to capture their distinct effects on consumer behavior (e.g., Haans & Gijsbrechts, 2011; Widdecke et al., 2023). While existing studies have provided valuable insights into the effects of promotional pricing across various contexts, notable gaps remain, particularly in how these strategies are applied across multi-format retail settings.

A number of studies (e.g., Raju, 1992; Shankar & Krishnamurthi, 1996; Widdecke et al., 2023) have explored the differential effects of regular prices and discounts, highlighting the distinct ways these components influence consumer behavior. However, fewer studies examine their simultaneous impact on brand performance across categories and store formats, offering limited insights into how these effects interact in practical retail scenarios. Although previous research (e.g., Pauwels et al. (2007), Haans & Gijbrechts (2011), and Gauri et al. (2017)) has examined the moderating roles of brand and category factors on the effectiveness of price and discount strategies, the direct and moderating influences of physical store formats on these pricing elasticities remain underexplored. Moreover, these studies often involve different retail chains and rely on third-party data, which can amplify observed differences, leaving empirically uncertain how these dynamics manifest within a single retail chain using data sourced directly from the retailer itself.

Our study addresses these gaps in three ways. First, we explicitly distinguish between the effects of regular prices and temporary discounts on brand sales, allowing us to identify asymmetries

Table 1: Overview of literature within promotional price effectiveness: Influence of store format, brand and category factor

Author(s) and Year	**Store Format**	**Brand Factor**	**Category Factor**	**Differential Effects of Price and Discounts**	**Retail Setting**	**Data Source**
Raju (1992)	No	No	Yes	Yes	One Store	Retailer
Shankar & Krishnamurthi (1996)	Yes	Yes	No	Yes	Different Chains	Third Party
Dhar, Hoch, & Kumar (2001)	No	No	Yes	Yes	Different Chains	Third Party
Nijs, Dekimpe, Steenkamp & Hanssens (2001)	No	No	Yes	Yes	Different Chains	Third Party
Ailawadi, Harlam, César & Trounce (2006)	No	Yes	Yes	Yes	Within Chains	Retailer
Pauwels, Srinivasan & Franses (2007)	No	Yes	Yes	No	Within Chains	Retailer
Haans & Gijsbrechts (2011)	Yes	No	Yes	Yes	Within and Across Chains	Third Party
Gauri, Ratchford, Pancras & Talukdar (2017)	No	Yes	Yes	No	Within Chains	Retailer
Jindal, Zhu, Chintagunta, & Dhar (2020)	Yes	Yes	No	No	Different Chains	Third Party
Datta, van Heerde, Dekimpe & Steenkamp (2022)	No	Yes	Yes	No	Different Chains	Third Party
Widdecke, Keller & Gedenk (2022)	Yes	Yes	Yes	Yes	Different Chains	Third Party
This Study	**Yes**	**Yes**	**Yes**	**Yes**	**Within Chains**	**Retailer**

in how consumers respond to price increases versus promotional discounts across store formats. Second, we examine these effects within a multi-level framework, where store formats not only directly influence pricing effectiveness but also moderate the role of brand- and category-level characteristics. Third, we leverage detailed loyalty-card data from a single retail chain to analyze multiple store formats within a unified system, thereby reducing heterogeneity in pricing policies, competitive environments, and customer composition. Together, these contributions provide a more precise understanding of how pricing strategies perform across retail formats and how their effectiveness depends on the interaction between format, brand, and category characteristics, offering actionable insights for retailers while extending the understanding of pricing effectiveness in multi-format retail settings.

2.2 Research Framework

Building on the identified theoretical and empirical research gaps, we propose a framework (Figure 1) that guides our study to investigate the differentiated impacts of regular prices and discounts on brand performance across various categories and store formats within the same retail chain.

In this study, promotional price effectiveness refers to the responsiveness of brand sales to changes in pricing instruments, captured through the estimated elasticities of regular prices and discounts. We distinguish between two related but conceptually different outcomes. First, we examine the effectiveness of discounts, measured as the sensitivity of sales to changes in temporary price reductions. Second, we analyze the differential effect between regular prices and discounts, which captures the relative magnitude of their impact on sales, rather than differences in sales levels. This distinction is important, as our focus is not on absolute sales differences, but on how strongly consumers respond to regular price changes compared to discounts across store formats.

The framework illustrates the role of store formats in influencing promotional price effectiveness and moderating how brand and category factors determine brand promotional price effectiveness (i.e., regular price elasticity and discount elasticity). It addresses the following research questions:
(1) How do store formats directly influence the differential effectiveness of regular prices and discounts and moderate the impact of brand factors and category factors on these differential effects?
(2) How do store formats directly influence discounts effectiveness and moderate the impact of brand factors and category factors on the effectiveness of discounts?

To answer these research questions, we explore evidence from academic research suggesting the potential impacts of store formats on promotional price effectiveness and their moderating role in brand and category factors in determining promotional pricing outcomes. Within the scope of brand and category factors, we differentiate between shared and distinct influences. Shared influences, such as discount depth (magnitude), breadth (coverage), are fundamental drivers of promotional effectiveness across both brand and category levels. These factors are directly influ-

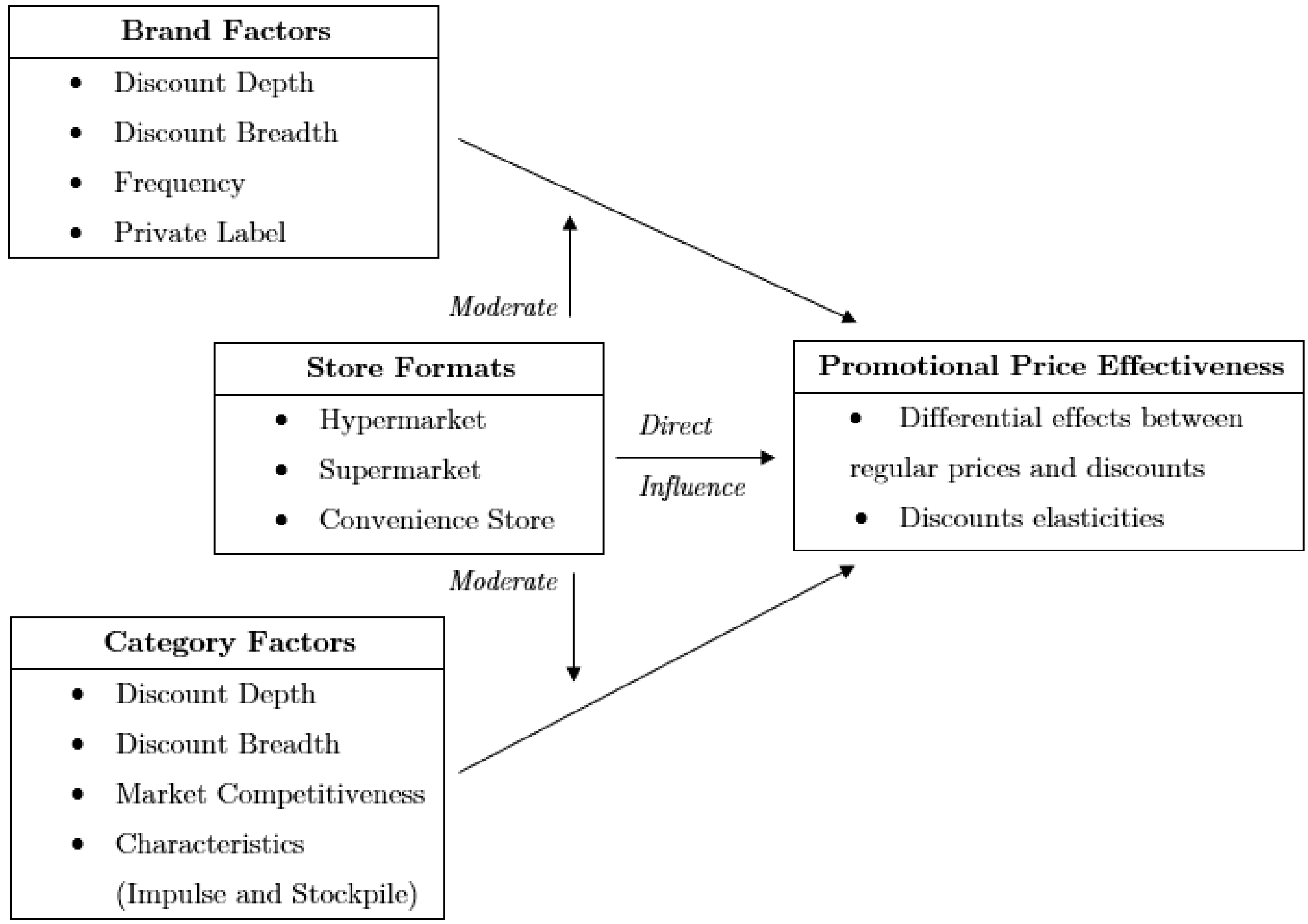

Figure 1: Research Framework

enced by promotional activities, as evidenced by studies like Gauri et al. (2017), which illustrate how broad-based discounting can significantly enhance store traffic and overall sales metrics. On the other hand, distinct factors such as frequency and private label status for brands, or competitiveness and product characteristics for categories, are well-established and commonly referenced determinants of price effectiveness (e.g. Widdecke et al., 2022).

2.2.1 Role of Store Formats in Influencing Regular Price and Discount Effectiveness

While most of the existing research focuses on the overall impact of final prices on sales, there is often a lack of detailed examination of how regular price adjustments and discounts specifically affect brand sales, and how these effects vary across different store formats. It has been documented that effectiveness of promotional pricing strategies often varies across store formats, reflecting differences in consumer shopping goals, preferences, and behaviors associated with each type of retail environment. For example, hypermarkets, supermarkets, and convenience stores within the same retail chain are designed to fulfill distinct needs such as bulk buying, regular grocery shopping, or immediate. These needs can influence how consumers respond to pricing strategies

(Koschmann & Isaac, 2018).

Previous research has highlighted these variations in promotional pricing effectiveness, yet much of the focus has been on broader measures such as incremental store sales (Ailawadi et al., 2006) or category sales (Haans and Gijsbrechts, 2011), rather than examining the direct effects of promotional pricing at the brand level and the differential impacts across various store formats. Moreover, the applicability of these broader findings to brand-level sales outcomes, which may exhibit distinct dynamics, remains uncertain. For instance, Jindal et al. (2020) demonstrate that promotional pricing tends to be most effective on brand sales in supermarkets, with a decreasing impact observed in drug stores, mass merchandisers, and convenience stores. This indicates that the effectiveness of price promotions can vary significantly across store types, influenced by the specific consumer shopping behaviors and expectations associated with each format.

The moderating role of store formats has often been examined indirectly, primarily in studies focusing on brand type and category-level factors influencing discount effectiveness. For instance, Widdecke et al. (2022) analyzed promotional effectiveness in supermarkets and discounters, including distinctions between private labels and national brands, but did not differentiate between the effects of regular price adjustments and temporary discounts. Similarly, Van Der Plas et al. (2024) demonstrated that price sensitivity varies between discounters and conventional retailers, particularly for national brands, yet their analysis did not extend to other store formats within the same retailer.

Although private labels and national brands have been studied extensively, limited attention has been given to how brand and category characteristics differently determine promotional price effectiveness across store formats within the same retail chain. The well-documented influence of product-specific factors on pricing effectiveness (e.g., Ailawadi et al., 2006; Gauri et al., 2017) highlights the importance of examining these dynamics across different store formats to provide a deeper understanding of promotional strategies. Store formats not only shape consumer expectations and shopping behaviors but also potentially influence how category characteristics and brand traits affect price-promotional outcomes on brand sales. Exploring these dynamics will illuminate how category- and brand-level factors moderate promotional effectiveness, paving the way for more tailored pricing strategies.

2.2.2 Brand Factors on Promotional Price Effectiveness

There are extensive studies examining how brand factors such as type (private label vs. national brands) and discount characteristics (depth, breadth, and frequency) influence promotional price effectiveness. However, less attention has been given to understanding how these factors separately affect regular prices and discounts across various store formats within the same retail chain.

The nature of the brand itself can determine the effectiveness of promotional pricing strategies. Whether a brand is a private label or a national brand significantly influences consumer price sensitivity. Private labels often elicit higher price sensitivity due to their positioning as value-for-money alternatives, unlike national brands that may benefit from stronger consumer loyalty and perceived quality (Kamakura & Russell, 1989; Krishna et al., 2002; Nijs et al., 2001). Recent research by Van Der Plas et al. (2024) highlights that national brands exhibit greater price sensitivity at hard discounters compared to conventional retailers. However, it is not yet clear whether these effects hold within the same retailer operating across multiple store formats. For example, national brands might experience varying degrees of price sensitivity depending on whether they are sold in hypermarkets, supermarkets, or convenience stores, where consumer expectations and shopping goals differ substantially.

In addition to brand positioning as a private label or a national brand, brand's discount characteristics such as depth, breadth and frequency can influence promotional effectiveness on brand sales. Previous research has shown that deeper and broader discounts are positively correlated with increased consumer purchases and consumption lift (Ailawadi et al., 2006). However, the frequency of discounts (i.e., how often a brand promotes its products) can become influential. Brands that frequently offer discounts may condition consumers to expect them, diminishing their sensitivity to both regular price changes and promotional offers over time (Pauwels et al., 2007; Srinivasan et al., 2004). While these patterns have been documented, it remains unclear whether the relationship between discount strategies and brand characteristics varies across store formats within a single retailer. For instance, private labels might rely heavily on frequent discounting in supermarkets, whereas national brands may need to adopt more tailored discount strategies in hypermarkets or convenience stores to align with format-specific consumer expectations.

2.2.3 Category Factors on Promotional Price Effectiveness

Previous literature has documented how category factors can affect promotional price effectiveness. However, there have been few attempts to understand how these category characteristics vary in their effects across different store formats within the same retail chain. Understanding the moderating role of store formats on category characteristics has significant managerial implications. To effectively implement promotional strategies, retailers have to consider how different category factors interact with store formats to better manage the impact of pricing on brand sales at the category level.

The attributes of product categories can shape consumer reactions to price promotion and influence the effectiveness of promotional prices. Prior research has identified storability and impulsiveness as key category traits that moderate the effects of promotional offers. Narasimhan et al. (1996) and Gauri et al. (2017) emphasize that in storable categories, where products can be easily stockpiled for future use, consumers are more likely to accelerate their purchases during promotions. This stockpiling behavior reflects a strategic response to capitalize on temporary price reductions, resulting in a substantial lift in brand sales during promotional periods, as consumers buy in bulk to save on future costs.

The impact of impulsiveness as a category characteristic on promotional price effectiveness, however, is mixed. Narasimhan et al. (1996) suggest that impulse-buy categories, characterized by spontaneous and unplanned purchases, often display unpredictable responses to price promotions. In these categories, consumer decisions are driven more by emotional triggers than rational price comparisons, potentially reducing the impact on brand sales. Nonetheless, Gauri et al. (2017) provide evidence that even in impulse-buy categories, promotional activities can lead to significant sales increases. Widdecke et al. (2022) further support this, indicating that while consumers in impulse categories may not respond strongly to the magnitude of the discount, they often exhibit heightened sensitivity to the mere presence of a promotional offer, leading to increased purchase behavior.

Apart from category static factors, the category's discount activities, including the category's discount depth and breadth, can shape consumer purchasing behavior. For instance, extensive and deep discounts across a category often lead to increased store traffic, higher sales per transaction, and improved profit margins (Dhar et al., 2001; Gauri et al., 2017). These outcomes suggest that category-wide discounting strategies can be a significant tool for retailers to stimulate demand and enhance overall sales performance.

The competitive landscape within a category is another important determinant of promotional price effectiveness. In highly competitive categories, where numerous brands vie for consumer attention, the impact of price promotions tends to be more pronounced, as consumers become more sensitive to price differences (Srinivasan et al., 2004; Datta et al., 2022). Conversely, in less competitive categories, consumers may exhibit weaker responses to promotions due to limited alternative choices and a reduced tendency to switch brands in response to price changes.

Despite these insights, it remains unclear how these effects vary across store formats within a single retail chain. For example, the effectiveness of promotions in storable categories may be enhanced in hypermarkets due to their larger scale and customer propensity for bulk buying, compared to convenience stores where impulse buying is more common and might not lead to the same volume of stockpiling. This variability suggests a need for retailers to understand format-specific promotional strategies that consider not just the general characteristics of product categories but also how these characteristics influence differently across store formats.

3 Data and Measures

3.1 Empirical Setting

Our study examines the differential effects of regular prices and discounts on brand performance across various product categories in different store formats using detailed scanner data from a Nordic grocery retailer. The dataset covers the period from January 1, 2014, to November 30, 2016 and comes from the retailer's loyalty membership database. The retailer operates under a model, where promotional strategies, such as discount campaigns, are centrally coordinated, but individual store managers retain flexibility in setting regular prices based on local market conditions. This setting creates a unique environment for analyzing pricing strategies, as it allows us to observe how regular prices and promotional discounts interact across different formats while accounting for store-specific implementation.

From the retailer business model, their system has three distinct store formats offered across regions, including a hypermarket, supermarket, and convenience store, each catering to different shopping goals and consumer preferences. The hypermarket is a large hypermarket situated on the outskirts of towns, offering an extensive selection of both grocery and non-grocery items at competitive prices. These stores cater to a larger customer base, focusing on convenience for car-borne shoppers, with a comprehensive assortment and shopping spaces. The supermarket features mid-sized supermarkets providing a broader range of products, emphasizing high service levels and catering to more comprehensive shopping needs. These stores are typically found in residential areas and aim to be the primary grocery destination for local consumers. The convenience stores consist of small, neighborhood stores offering convenient and localized shopping experiences, often located close to residential areas or workplaces. These stores have a limited but tailored assortment focused on everyday essentials and fresh produce, with a strong emphasis on service. By studying these three store formats, we capture a diverse range of consumer behaviors and retail strategies within the same grocery chain.

Store formats in this setting represent standardized retail concepts defined by the retailer, implying structural differences in assortment, pricing strategies, and shopping objectives. While differences in pricing effectiveness could, in principle, arise from store-specific characteristics such as size, assortment, or location, our empirical setting is designed to minimize such variation. We focus on three stores within the same retail chain and geographic area, serving a largely overlapping customer base. As such, format provides a clean and managerially relevant construct to study how consumers respond to pricing and promotions. At the same time, we acknowledge that some residual within-format heterogeneity may remain, and our results should be interpreted as reflecting format-level differences rather than purely store-specific effects.

3.2 Data

3.2.1 Selected Stores

The dataset from the retailer's loyalty membership database includes detailed purchasing information from customers selected through the retailer's loyalty program. It tracks comprehensive shopping behavior, covering who made purchases, the items bought, the timing and quantities of purchases, and the prices paid across all stores within the same retailer chain nationwide. This extensive dataset offers detailed information for analyzing the distinct effects of regular prices and discounts on brand sales across store formats.

Despite the detailed coverage, data at the individual store level can be sparse due to the wide distribution of customer shopping across various locations. This is because the dataset is customer-centric, meaning that store-level observations are inferred from individual shopping behavior rather than directly observed. As a result, many stores are visited only sporadically by the sampled customers, leading to highly sparse store–brand–week observations. To minimize potential geographic and economic disparities and to ensure the quality and continuity of data for our analysis, we focus our analysis on three specific stores, each representing one of the formats (hypermarket, supermarket, and convenience store) located within the same city. These stores were selected because they were frequently visited by a consistent subset of customers from the loyalty program, thus providing a reliable dataset for observing the effects of pricing strategies. Including all stores would introduce substantial noise and discontinuities in the time series, which would complicate the estimation of pricing elasticities that require sufficiently dense and consistent observations over time. Figure 2 illustrates the locations and relative distances of the three store formats, highlighting their proximity within the same area. This geographic focus helps control for external factors, allowing us to isolate the effects of regular pricing and discount strategies across different retail environments.

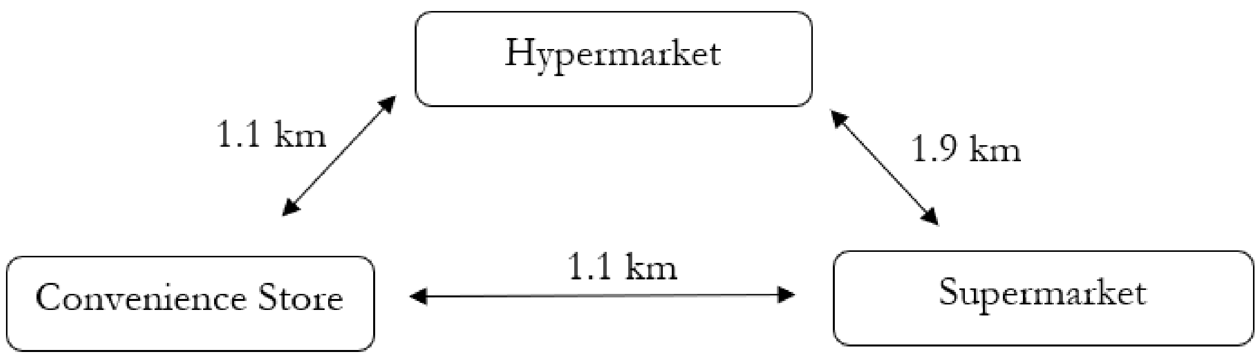

Figure 2: Simple illustration of selected store formats offered by the same retailer chain

We present our initial dataset regarding our empirical setting in Table 2. The data from the retailer's loyalty membership database includes randomly selected 5,468 customers who made total 897,743 grocery shopping trips. Our selected store formats (i.e., hypermarket, supermarket, and convenience store) are significantly different from one another in characteristics and customers' shopping behaviors per trip. Consistent with the described retailer's business model, the three store formats differ significantly in size, assortment, and customer shopping behavior, reflecting their distinct roles within the same retail chain.

The hypermarket, as the largest format (4,252 m^2), offers the widest product assortment, with 1,517 unique brands and 17,400 different products. Shoppers in hypermarkets exhibit the highest average spending per trip (246 SEK), with a broader variety of items purchased, including an average of 10 unique products and 7 brands per visit. The visiting frequency (0.56 trips per week)

Table 2: Customers and their spending per trip by store format

	Hypermarket	**Supermarket**	**Convenience**
Size (m^2)	4,252	2,155	497
Total unique brands available	1,517	1,388	680
Total unique products available	17,400	16,026	7,325
Total visitors	5,419	4,499	3,367
Total visits	461,878	314,164	121,701
Avg spending per trip	246	187	110
Avg category per trip	7	6	3
Avg trip per week	0.56	0.46	0.24
Avg unique products per trip	10	7	4
Avg unique brands per trip	7	6	3

Notes: Reported across 897,743 shopping trips from 5,468 customers from the same demographic area visiting these store formats. The table reported the size (square meters), the number of brands and products purchased (units), and the average customers' shopping behaviors in each format per visit. The unit of the spending is in the local currency (SEK).

aligns closely with supermarkets, reflecting their shared role as primary grocery destinations. This trend is consistent with recent survey findings (Statista, 2024), which highlight hypermarkets and supermarkets as top choices among Swedish consumers. Most visits occur on Fridays, likely driven by preparations for the weekend, while Sundays have the least traffic (Figure 3), possibly due to cultural preferences for relaxation and family time.

The supermarket format, representing a middle ground, features a smaller size (2,155 m^2) and slightly reduced assortment, offering 1,388 unique brands and 16,026 products. The average spending per trip is lower (187 SEK), and customers typically purchase around 7 unique products and 6 brands. This format is strategically positioned to serve local communities as the primary grocery shopping destination, attracting customers with a balanced assortment and a convenient shopping experience. The visiting frequency (0.46 trips per week) and similar visit patterns to hypermarkets (Figure 3) suggest that consumers rely on these formats for their main grocery needs, particularly on Fridays.

The convenience store, being the smallest format (497 m^2), focuses on immediate, essential purchases with a limited but targeted assortment of 680 unique brands and 7,325 products. Customers at convenience stores exhibit the lowest average spending per trip (110 SEK), with fewer items (4 unique products) and brands (3 brands) purchased. The trip frequency (0.24 trips per week) is also the lowest, indicating that consumers use this format primarily for quick, unplanned purchases or emergency needs, aligning with the retailer's strategy of offering accessible, neighborhood-based shopping options. In contrast, convenience stores display a more balanced visit distribution throughout the week, with a noticeable peak on Tuesdays (Figure 3). This suggests that consumers rely on convenience stores for unplanned, urgent purchases or smaller trips. The lower visit percentage on weekends (especially Saturdays) further supports the idea that consumers prefer to visit larger formats when planning their weekly shopping.

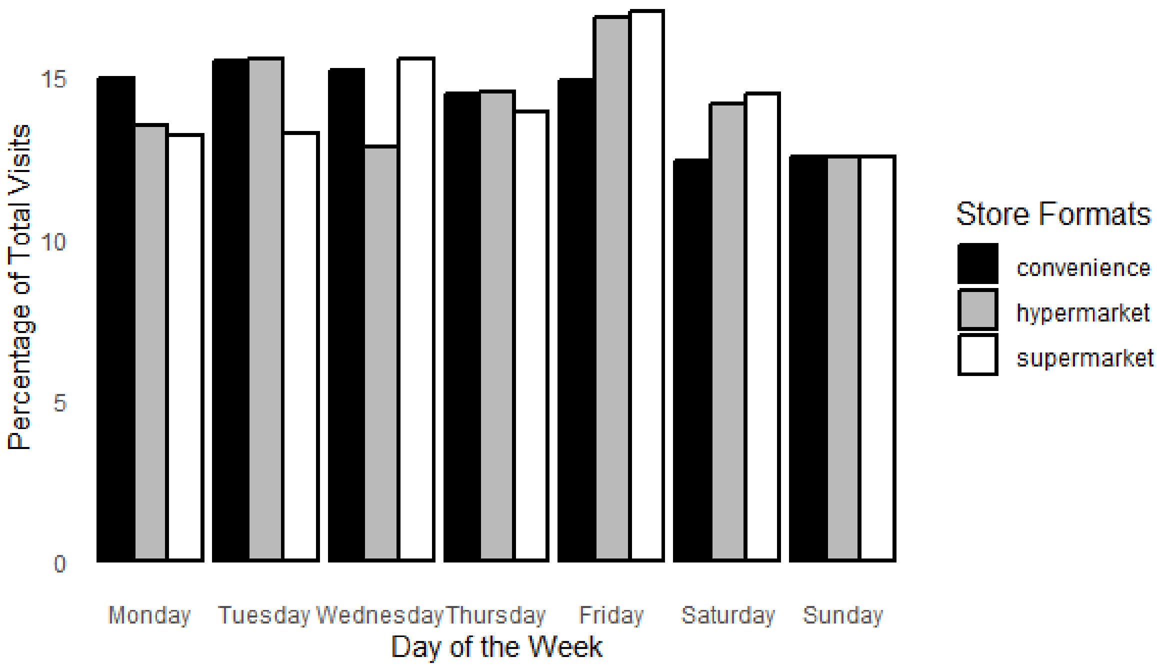

Figure 3: Percentage of Store Format Visits By Day-of-the-Week

Overall, the differences across store formats reflect the retailer's franchise model, where each format is designed to cater to distinct shopping goals and customer preferences. These format distinctions are consistent with the findings by Jindal et al. (2020), who demonstrated that supermarkets and mass merchandisers are frequented for major trips associated with higher spending and broader category purchases. In contrast, convenience stores primarily serve unplanned, urgent trips characterized by smaller baskets and fewer items. Their analysis highlights that consumer responses to marketing-mix variables differ across retail formats, with brand shares being more sensitive to line length changes at convenience stores due to their limited assortments and less frequent trips. This alignment supports the empirical setting of our study, providing an appropriate sample for analyzing how regular pricing and discounts influence brand sales differently across hypermarkets, supermarkets, and convenience stores.

3.2.2 Selected Brand and Category

Given that our dataset comes from the retailer's loyalty membership database, the granularity it offers in terms of customer purchases is significantly more detailed than what is typically available through third-party data (van Heerde and Dekimpe, 2024). Such data allows us to track the same customers who shop across different formats, offering insights into cross-format shopping patterns. However, this detailed level of customer purchasing information from loyalty membership data means we have more limited visibility into the full range of brands offered across all store formats.

To address this limitation, we have to carefully select specific brands and categories for analysis that are consistently available across selected hypermarket, supermarket, and convenience store. This approach ensures we capture a sufficient dynamic of pricing strategies and brand performance within each selected store format.

While this method allows for a focused analysis of selected brands and categories, it introduces a potential drawback: the loss of information regarding products and brands that are uniquely tailored or primarily available in specific store formats. Different store formats are often designed to cater to varying consumer needs and preferences, which may mean that some SKUs or product categories are not uniformly available across formats. This selective availability can lead to gaps in our understanding of category and product performance, especially for those items that do not meet our selection criteria due to their limited distribution. Despite this limitation, our focus remains on analyzing the brand performance and pricing strategies effectively within the confines of the available and consistent data across the chosen formats.

First of all, we focus on frequently purchased products (Stock Keeping Unit-level) for each category. We initially selected the top five sizes of each product category that demonstrated high sales volume on a weekly basis. This step is to ensure that comparisons across brands were made between like-for-like products, thereby avoiding inconsistencies that arise from comparing different package sizes. Second, we identify the most frequently purchased products within each category, specifically certain products that are bought at least once every two weeks. These products must have consecutive non-zero sales throughout the entire study period (i.e., 153 weeks), in line with Widdecke et al. (2022), to reduce the risk of noise from sporadic or irregular sales patterns. From the product characteristics identified in the first two steps, we then select those that match in size and have similar product descriptions. This is done to ensure that our comparisons are not only consistent in terms of physical dimensions but also in terms of product description across different brands.

We further focus on frequently purchased brands. We narrow down our selection to include only those brands whose products described earlier are purchased every two weeks across at least two different store formats. This step helps ensure that our analysis can reliably measure how the same brands perform under different format conditions. Lastly, we retain only categories that maintain at least three brands, which are purchased every two weeks across at least two different store formats. This category selection is not only for maintaining sufficient sample sizes for second-stage analysis but also for ensuring that our analysis can capture a range of consumer responses to different brands within the same category across multiple store formats. Moreover, this selection process facilitates interpretation from a retailer's perspective, providing actionable insights into which products and brands could be strategically adjusted in price to enhance sales performance.

Our sample selection process requires scrutinized consideration due to the dataset's characteristics and the multi-format retail environment. While our data does not directly capture store-level data, this information can be inferred from customer transactions. Although inferring store-level data from customer transactions may introduce some variability, this is a well-accepted approach in multi-format retail analysis (e.g., Widdecke et al., 2022; Van Der Plas et al., 2024). As a result of these carefully chosen selection criteria, we retained a dataset used for the analysis comprising 233,456 weekly shopping trips from 5,384 customers. This final sample includes 55 unique brands across 14 frequently purchased categories[1] (Table 3). While our final dataset focuses on 55 unique brands across 14 categories for examining price and discount elasticities, it is important to note that there are total 66 unique brands whose products frequently purchased. These 11 additional brands are used to construct competitor's price information within the analysis. However, they do not meet the criteria for inclusion as focal brands in our elasticity study because they are not purchased consistently across more than two store formats.

Table 3: Focal Categories and Brands Included in This Study

Store Format	Hypermarket	Supermarket	Convenience
Total Unique Visitors	5,193	4,176	2,743
Total Sales	16,853,451	8,557,788	1,525,896
Number of Focal Category	14	14	13
Number of Focal Brand: Baking	3	3	1
Number of Focal Brand: Berries	3	3	2
Number of Focal Brand: Bottled water	3	3	3
Number of Focal Brand: Butter	9	9	5
Number of Focal Brand: Cereals	5	5	-
Number of Focal Brand: Chips	3	3	3
Number of Focal Brand: Chocolate	4	4	2
Number of Focal Brand: Cream	6	6	4
Number of Focal Brand: Fruit Juice	10	10	3
Number of Focal Brand: Ice cream	5	5	3
Number of Focal Brand: Milk	3	3	1
Number of Focal Brand: Pasta	5	5	3
Number of Focal Brand: Soft drinks	8	8	6
Number of Focal Brand: Toiletries	4	4	2

Notes: The table reports sales value in local currency.

3.2.3 Example of Selection Process

To demonstrate our selection process, we provide an example of how category of soft drinks is selected. In our dataset, soft drinks are purchased by customers in our loyalty program across a variety of store formats and locations, including occasional purchases in stores outside of their usual shopping area, such as during trips to the capital city. These variations introduce significant price disparities and shopping behaviors, complicating the analysis of pricing strategies and brand

[1]These categories include: Baking ingredients, Berries, Bottled water, Butter, Cereals, Chips & Salty snacks, Chocolate, Cream, Fruit Juice, Ice cream, Milk, Pasta, Soft drinks, Toiletries.

performance. To mitigate these complications, we focused on selecting stores located in the city where our customers reside and shop regularly so we could compare shopping behaviors across consistent geographic and economic conditions and across different store formats—hypermarkets, supermarkets, and convenience stores.

The variety of packaging for soft drinks ranges widely, from large 3-liter bottles to packs of 20 small cans. Such diversity in product sizes and types can skew straightforward comparisons. Therefore, to maintain comparability across store formats, we selected specific product sizes that are commonly available and frequently purchased across all three store formats. We chose to focus on sugary carbonated drinks, a popular sub-category within soft drinks, and standardized the product sizes for analysis: 330ml cans, 500ml normal bottles, and 1500ml big bottles. These sizes are prevalent across all formats, ensuring a fair comparison. Additionally, to facilitate more accurate comparisons and analysis, all package sizes were converted into units of 100ml. This conversion allows us to normalize the data, making it possible to compare pricing and quantity across different package sizes effectively.

Observing customer purchases over 153 weeks (i.e., 77 biweekly times), seasonal and festive-specific products, such as specialty soft drinks that are heavily purchased during specific periods (e.g., Christmas), pose a challenge. These products, while significant in volume during certain times, do not represent regular purchasing patterns throughout the year. Including such items could distort the analysis of regular price sensitivity and discount effectiveness. Consequently, we exclude products that are not consistently purchased throughout the study period of 153 weeks. By doing so, we focus on those brands and products that exhibit stable sales, providing a more accurate reflection of ongoing consumer behavior and brand performance.

3.3 Variable Operationalization

3.3.1 First-stage variables

In the first-stage analysis, we quantify the effects of regular prices and discounts on brand performance (i.e., brand sales in unit) by estimating their elasticities. We aggregate customer purchase data from the retailer's loyalty membership database to the weekly brand level for each store format. To ensure accuracy in our estimates, we aggregate customer purchase data from the retailer's loyalty membership database to the weekly brand level for each store format. This aggregation is made possible by our selection process, which ensures that the products and brands included in our dataset are consistently purchased across all selected store formats, thereby providing reliable measures of aggregated brands' prices and performance data. Table 5, Panel A provides details about the operationalization of the variables in the first-stage analysis.

The dependent variable in our analysis is the weekly brand volume sales. To estimate the elasticities, we employ the sales-response model proposed by Pauwel et al. (2007) having brand price (i.e., regular price and discounts), competitor brands price, and non-price promotion as independent variables. Since we are interested in changes of sales across format, we control for additional variations, apart from the seasonality, that might affect sales independently of price changes.

First, we include category visits, the number of customers in the loyalty membership database visiting the selected store in a given period to buy products in a given category. This variable accounts for the average traffic each category receives across different store formats. Higher traffic can often lead to variations in sales figures, which are independent of our primary variables of interest, such as price changes.

Second, we include average share of trip types. Following the methodology outlined by Jindal et al. (2020), we construct trip types using initial transaction data including baskets of all brands and categories. We analyze characteristics of the basket or receipt as follows. Trips were categorized as 'major' if their spending levels, number of items purchased, or number of product categories exceeded the median values for each customer. We then employ K-means clustering to further classify the non-major trips into 'fill-in' and 'unplanned minor' trips based on the following basket characteristics: average price per item, number of items and categories purchased, the number of days since the last trip to account for intershopping times [2]. We report the average trip characteristics per trip per household by their trip type in in table 4. To address potential issues of collinearity—which could arise because the shares of trip types sum to one—we selectively include only the share of major trips and the share of unplanned minor trips in our analysis.

Table 4: Average Statistics (Per Trip) by Trip Type

	Major	**Fill-in**	**Unplanned Minor**
Average Price Per Item	25	17	24
Average Total Items Purchased	16	5	2
Average Total Categories Purchased	9	4	2

Notes: Reported average trip characteristics per trip per household across 904,619 trips made by 5,435 households from anuary 1, 2014, to November 30, 2016.

The rationale for including trip type as an additional control variable is to control for observed potential relation between shopping trip characteristics and store format, and self-selection bias. The empirical setting described in section 3.2.1 implies distinct patterns related to each store format—such as average spending per trip, the typical number of categories per trip, and the assortment of products purchased. These patterns suggest that trip characteristics might vary systematically across formats, influencing sales independently of price changes. These observed patterns correspond to the trip type classifications outlined by Jindal et al. (2020). Controlling for trip types helps mitigate potential self-selection bias, where customers in the loyalty program may choose a particular store format based on specific needs or shopping goals. Without accounting

[2]Following Jindal et al. (2020), we standardize these variables at the house-hold level to ensure that trips are clustered on the basis of within-household differences.

for these trip types, our analysis could incorrectly attribute differences in sales performance to pricing strategies rather than to the underlying shopping intent and store format preference.

3.3.2 Second-stage variables

In the second-stage analysis, we aim to quantify systemic differences in the effectiveness of regular pricing and discounts across different store formats, and explore the role of store formats in influencing how brand and category factors determine these price elasticities. According to our research framework, brand factors include brand's discount depth, brand's discount breadth, brand's discount frequency and whether the brand is a private label. Category factors include category's discount depth, category's discount breadth, category's market competitiveness. For category characteristics (i.e., impulsiveness and storability), we use the scale from the data employed in Gauri et al. (2017) and a consumer survey conducted in 2020 by a marketing research company in The Nordics. Table 5, Panel B provides details about the operationalization of the variables in this second-stage analysis.

3.4 Descriptives

Table 6 presents descriptive statistics for our variables across different store formats within our dataset. The hypermarket generally exhibits the lowest average regular prices and the greatest standard deviations in prices, indicating a broader range of prices available compared to other formats. In terms of discounts, while the average discount per-unit discounts across brands and categories tends to be relatively low, the hypermarket still displays relatively highest discount as well as the standard deviations. Conversely, the convenience store tends to have higher average regular prices across many categories, which could be attributed to their focus on convenience and immediate accessibility. Additionally, the convenience store engages less frequently in discount activities at both brand and category level. The supermarket, positioning between hypermarkets and convenience stores, shows a moderate level of pricing and discount activity. Overall, the behavior of pricing and discounting of our selected stores, brands and formats aligns with the retailer business model.

In addition to pricing and discounting which are key interests, our study controls category visits and the distribution of trip types across store formats. The hypermarket tends to attract a larger volume of visits and a higher proportion of major shopping trips. This aligns with their role as destinations for extensive, planned shopping activities. On the other hand, the convenience store records a greater share of unplanned trips, highlighting their role in catering to immediate, spontaneous needs. The supermarket, with their moderate visitor counts, frequently host both planned and unplanned trips (and fill-in trips), illustrating their versatile shopping environment.

Table 5: Variable Operationalization

Variable	**Operationalization and Description**
Panel A: Variables in First-Stage Sales-Response Model	
$\text{Sales}_{ijk,t}$	Unit sales for brand i in category j in store format k in week t.
$\text{RegPrice}_{ijk,t}$	Regular price of brand i in category j in store format k in week t, computed as a sales-weighted average across its SKUs sold in period t, expressed in local currency.
$\text{Discount}_{ijk,t}$	Discounts offered of brand i in category j in store format k in week t, computed as a sales-weighted average across its SKUs sold in period t, expressed in local currency.
$\text{FD}_{ijk,t}$	Intensity of feature and/or display support for brand i in category j in store format k in week t, computed as a sales-weighted average across its SKUs sold in period t, expressed in number range from 0 to 1.
$\text{CompPrice}_{ijk,t}$	Sales-weighted average of brands i's competing brands' ($\neq i$) average price paid in category j in store format k in week t.
$\text{CVisitors}_{jk,t}$	Total number of customers made purchase in category j in store format k in week t.
$\text{ShareMajor}_{ijk,t}$	Average share of Major Shopping Trip made for brand i in category j in store format k in week t.
$\text{ShareUnplan}_{ijk,t}$	Average share of Unplanned Shopping Trip made for brand i in category j in store format k in week t
Panel B: Variables in Second-Stage Regression	
Brand Factors	
BDepth_{ijk}	Average weekly ratio of discounts to regular prices of brand i in category j in store format k.
BBreadth_{ijk}	Average weekly ratio of number of SKUs getting promoted to total number of SKUs of brand i in category j in store format k.
BFreq_{ijk}	Ratio of number of weeks in which negative price-promotion shocks are at least 5% of the brand i's regular price to total number of weeks in the study in category j in store format k
PL_{ij}	An indication variable equal to 1 if brand i in category j is owned by the retailer.
Category Factors	
CDepth_{jk}	Average weekly ratio of discounts to regular prices of category j in store format k.
CBreadth_{jk}	Average weekly ratio of number of SKUs getting promoted to total number of SKUs of category j in store format k.
CComp_{jk}	Variance in shares across brand of category j in store format k
Impulse_{j}	An indication variable equal to 1 if category j is impulsive (scale reported by Gauri et al. (2017) and the survey
Storable_{j}	An indication variable equal to 1 if category j is storable (scale reported by Gauri et al. (2017) and the survey

Table 6: Descriptives Statistics

	Hypermarket	Supermarket	Convenience
Variables in First Stage: RegPrice and Discount			
Brand RegPrice: Baking	10.11 (1.32)	10.95 (1.18)	10.60 (1.76)
Brand RegPrice: Berries	6.68 (1.88)	7.26 (2.28)	7.03 (2.52)
Brand RegPrice: Bottled water	0.83 (0.08)	0.88 (0.14)	1.05 (0.25)
Brand RegPrice: Butter	4.02 (1.21)	4.23 (1.24)	4.17 (1.44)
Brand RegPrice: Cereals	6.41 (1.45)	6.60 (1.45)	-
Brand RegPrice: Chips & Salty snacks	7.02 (0.71)	6.79 (1.18)	7.84 (1.29)
Brand RegPrice: Chocolate	13.60 (5.16)	15.43 (4.18)	14.31 (1.62)
Brand RegPrice: Cream	3.81 (0.44)	3.83 (0.38)	4.61 (1.04)
Brand RegPrice: Fruit Juice	1.66 (0.70)	1.72 (0.71)	1.83 (0.28)
Brand RegPrice: Ice cream	3.27 (3.23)	3.65 (3.14)	2.27 (0.59)
Brand RegPrice: Milk	0.90 (0.10)	0.96 (0.14)	1.33 (0.53)
Brand RegPrice: Pasta	1.81 (0.39)	1.75 (0.43)	1.96 (0.36)
Brand RegPrice: Soft drinks	1.14 (0.17)	1.35 (0.30)	1.45 (0.39)
Brand RegPrice: Toiletries	25.38 (7.79)	26.16 (7.43)	27.76 (8.71)
Brand Discount: Baking	0.14 (0.22)	0.11 (0.18)	0.20 (0.37)
Brand Discount: Berries	0.19 (0.37)	0.19 (0.40)	0.26 (0.62)
Brand Discount: Bottled water	0.02 (0.03)	0.02 (0.03)	0.03 (0.03)
Brand Discount: Butter	0.08 (0.16)	0.10 (0.22)	0.07 (0.21)
Brand Discount: Cereals	0.08 (0.11)	0.06 (0.12)	-
Brand Discount: Chips & Salty snacks	0.34 (0.49)	0.18 (0.23)	0.11 (0.24)
Brand Discount: Chocolate	1.04 (1.55)	0.46 (1.22)	0.39 (0.92)
Brand Discount: Cream	0.10 (0.15)	0.12 (0.18)	0.10 (0.27)
Brand Discount: Fruit Juice	0.05 (0.07)	0.04 (0.05)	0.04 (0.07)
Brand Discount: Ice cream	0.06 (0.14)	0.11 (0.28)	0.09 (0.23)
Brand Discount: Milk	0.01 (0.01)	0.01 (0.01)	0.00 (0.01)
Brand Discount: Pasta	0.07 (0.12)	0.05 (0.13)	0.07 (0.18)
Brand Discount: Soft drinks	0.08 (0.06)	0.08 (0.07)	0.09 (0.12)
Brand Discount: Toiletries	0.76 (1.23)	0.63 (1.24)	0.62 (1.73)
Variables in First Stage: Category Visits and Share of Trip Types			
CVisitors	875.60 (656.87)	437.60 (326.57)	104.76 (80.12)
ShareMajor	0.89 (0.07)	0.84 (0.10)	0.51 (0.17)
ShareUnplan	0.03 (0.04)	0.04 (0.05)	0.18 (0.13)
Variables in Second Stage: Brand Factors			
BDepth	0.033 (0.024)	0.029 (0.020)	0.030 (0.020)
BBreadth	0.041 (0.037)	0.039 (0.033)	0.030 (0.032)
BFreq	0.209 (0.215)	0.187 (0.177)	0.306 (0.218)
Variables in Second Stage: Category Factor			
CDepth	0.033 (0.027)	0.025 (0.015)	0.021 (0.019)
CBreadth	0.232 (0.093)	0.212 (0.093)	0.120 (0.048)
CComp	0.041 (0.041)	0.0035 (0.041)	0.040 (0.046)

Notes: The table reports the mean price per unit (SD) across 180 brands and their factors, as well as the mean variables of category factors (SD) across 14 categories, aggregated over 233,456 shopping trips by 5,384 customers in the same demographic area. These figures are averages calculated over 77 biweekly periods.

4 Model-free evidence

We start by providing model-free evidence for our research framework in proposing the role of store format in influencing how brand and category factors determine brand promotional price effectiveness (i.e., regular price elasticity and discount elasticity). In this section, we focus on the bottled water category as an example for model-free evidence. We select one national brand and one private label to investigate the relationship between changes in regular price and discount with respect to unit sales across different store formats. Figures 4 and 5 visually depict how sales volume responds to changes in regular prices and discounts for the national brand and private label respectively.

The plots illustrate that the relationship between promotional price (i.e., regular prices and discounts) changes and sales volume varies not only across formats but also between the national brand and the private label. Notably, for a given change in regular price, the slope of the response curve differs across store formats, indicating that the impact of price adjustments on sales volume is potentially influenced by the type of store. This variation suggests that consumer sensitivity to price changes might be different in the hypermarket compared to the supermarket and the convenience store. When comparing the response to price changes between the national brand and the private label within the same format, we observe distinct patterns. The national brand tends to show a more pronounced change in sales volume for similar price changes compared to the private label. However, in the convenience store, the effect of price changes and discounts for the national brand appears to be symmetrical, suggesting that reductions in regular price and increases in discounts have similarly proportional effects on sales volume.

In principle, if changes in regular price and discounts had symmetrical effects on sales, the relationship depicted would mirror each other. This means that a decrease in regular price would have a comparable impact to an equivalent increase in discount. For example, in convenience stores, one might expect that price reductions and increases in discounts would have similar effects on sales volumes. However, the figures illustrate that this is not always the case; for instance, the relationship shown in figure 5 for the private label in hypermarkets reveals that sales volumes increase significantly with discounts but do not decrease equivalently with similar increases in regular prices. This asymmetry implies that consumers may perceive and react to price decreases and discount increases differently. These empirical observations suggest that store formats influence how pricing strategies affect sales.

5 Method

In this study, we examine the differential effects between regular prices and discounts on brand performances offered across different categories in different store formats. We also investigate the potential determinants of discount effectiveness from brand and category factors and compare them

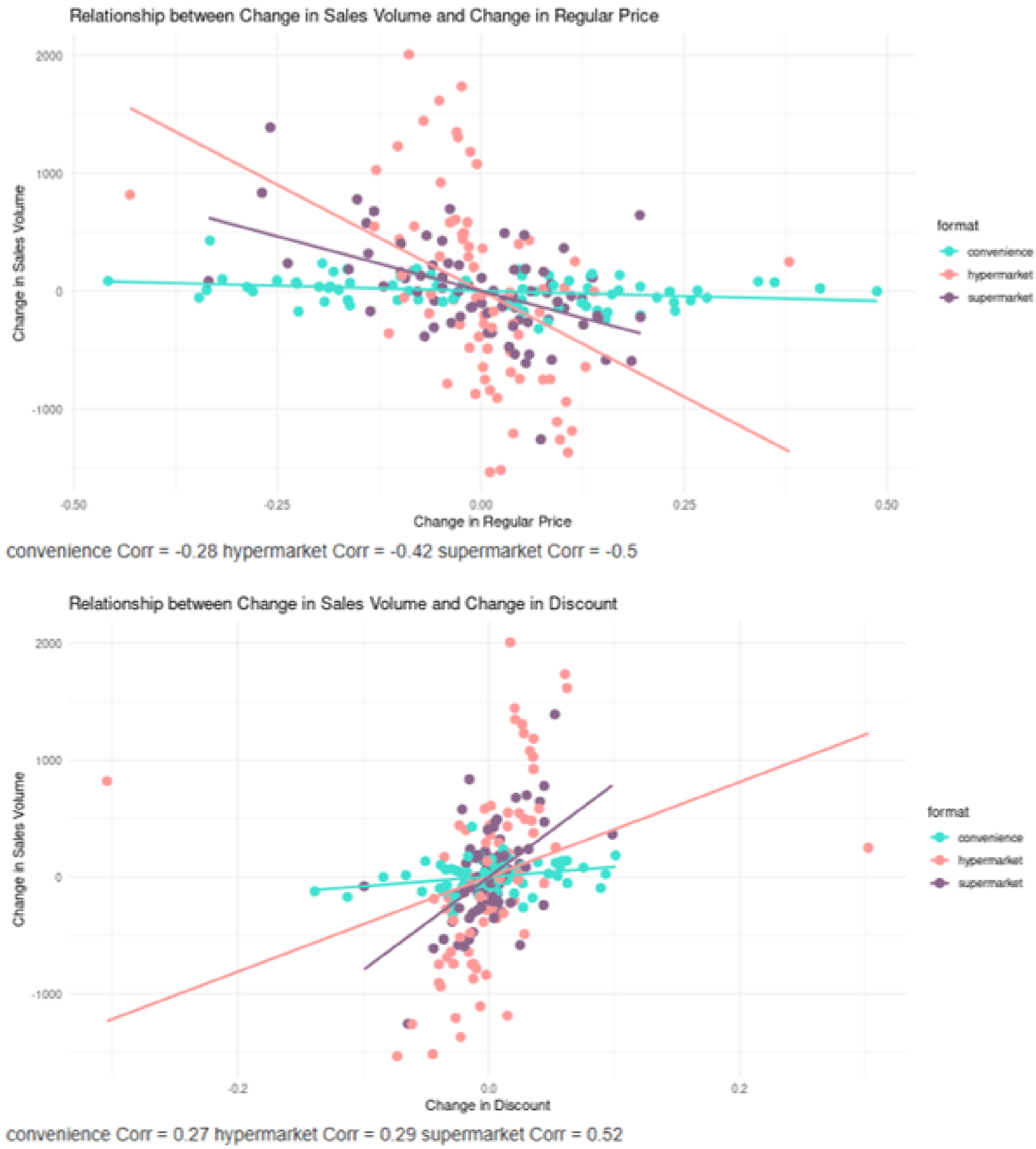

Figure 4: A scatter plot of time series of the changes in sales (100ml) of bottled water with respect to the changes in regular price and discount (per unit) and their correlations for national brand across formats

across store formats. In line with previous studies (e.g., Datta et al., 2022), we first estimate regular price and discount elasticities across 55 brands using sales-response models. We then estimate a regression with respect to (1) the differences of elasticities and (2) discount elasticities using brand and category factors and store formats (i.e., hypermarket, supermarket and convenience store) as moderators.

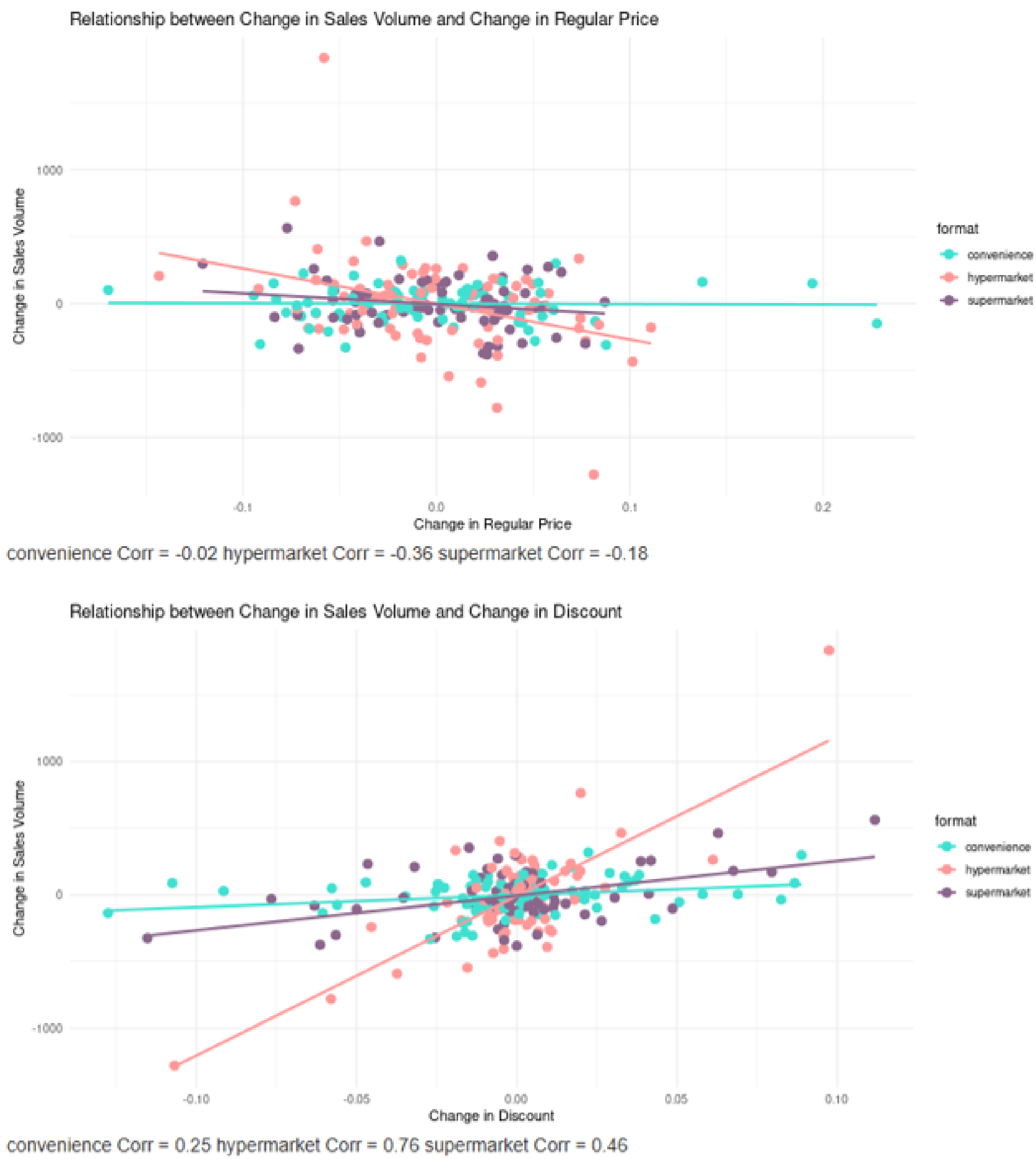

Figure 5: A scatter plot of time series of the changes in sales (100ml) of bottled water with respect to the changes in regular price and discount (per unit) and their correlations for private label across formats

5.1 Sales-Response Model Specification

In order to quantify the effect of regular prices and discounts, we adopt an error correction specification. This specification allows us to estimate the short-run elasticity of price and discount while accounting for potential long-run equilibrium relationships (Pauwels et al., 2007). In our sales response model specification, we use linear model to avoid the potential aggregation bias that might happen in other specifications such as log-log specification (Datta et al., 2022). Moreover, the linear model provides straightforward interpretation of the coefficients. For instance, the coefficient of the discount can be interpreted such that a change of one unit in the local currency of the discount leads to an estimated change in unit sales. This interpretation is direct and does not require further transformation, making it easy to understand the impact of price changes and discounts on sales.

Adopting an error-correction specification, the model for brand sales i in category j in store format k is:

$$\begin{aligned}(1)\quad \Delta S_{ijk,t} = {} & \beta_0 + \beta_{1ijk}\Delta RegPrice_{ijk,t} + \beta_{2ijk}\Delta Discount_{ijk,t} \\ & + \beta_{3ijk}\Delta CompPrice_{ijk,t} + \beta_{4ijk}FD_{ijk,t} \\ & + \gamma_{ijk}[S_{ijk,t-1} - \alpha_{0ijk} - \alpha_{1ijk}RegPrice_{ijk,t-1} - \alpha_{2ijk}Discount_{ijk,t-1}] \\ & + \beta_{5ijk}CVisitors_{jk,t} + \beta_{6ijk}ShareMajor_{ijk,t} + \beta_{7ijk}ShareUnplan_{ijk,t} \\ & + \beta_{8ijk}Holiday_t + \beta_{9ijk}Week_t + \beta_{11ijk}Year_t \\ & + \beta_{11ijk}CopulaRegPrice_{ijk,t} + \beta_{12ijk}CopulaDiscount_{ijk,t} + \varepsilon_{ijk}\end{aligned}$$

where $\Delta X_t = X_t - X_{t-1}$ indicates the first difference of variable X, emphasizing changes rather than levels to address potential non-stationarity in time series data. The coefficients $\hat{\beta}_{1ijk}$, $\hat{\beta}_{2ijk}$, $\hat{\beta}_{3ijk}$ and $\hat{\beta}_{4ijk}$ capture the immediate impact of changes in regular price, discount, competitors' prices, and brand's feature and display, respectively, on the sales volume. The error-correction term represented by γ_{ijk} measures how quickly sales revert to equilibrium following a change in regular prices and discounts, highlighting the dynamic adjustment process. Additional controls including category-specific visits in each store format ($CVisitors_{jk,t}$) and average shares of shopping trip types ($ShareMajor_{ijk,t}$,$ShareUnplan_{ijk,t}$) are introduced in the model. Seasonal effects are controlled through dummy variables for holidays, weeks, and years to ensure the model captures any systematic variations specific to these periods. In addition to variables of interest, we added the Gaussian copula for $RegPrice_{ijk,t}$ and $Discount_{ijk,t}$ to address the potential presence of endogeneity (e.g., endogeneity caused by idiosyncratic changes in consumer preferences for brands) (Datta et al., 2022).

In this error-correction model, we obtain short-term effects for regular price ($\hat{\beta}_{1ijk}$) and discount ($\hat{\beta}_{2ijk}$) on sales for each brand. These coefficients represent the change in sales units for a one-unit change in price or discount (in local currency). However, given that different categories have different units of sales, we control for scale differences by converting the obtained parameters ($\hat{\beta}_{1ijk}$, $\hat{\beta}_{2ijk}$) to unit-free elasticities. This is done by normalizing the estimated parameters with the ratio of the sample mean[3] of promotional price of each brand (i.e., regular price and discount)

[3]Since we observe irregular drops of regular prices in certain products, we use the sample median instead of mean to avoid the extreme values in time-series.

to sample mean of sales of each brand (Srinivasan et al., 2004; Datta et al., 2022). Thus we get,

$$\hat{\eta}_{1ijk} = \hat{\beta}_{1ijk} * \frac{(\overline{RegPrice}_{ijk})}{(\overline{Sales}_{ijk})}$$

$$\hat{\eta}_{2ijk} = \hat{\beta}_{2ijk} * \frac{(\overline{Discount}_{ijk})}{(\overline{Sales}_{ijk})}$$

where $\hat{\eta}_{1ijk}$ is the estimated regular price elasticity at mean and $\hat{\eta}_{2ijk}$ is the estimated discount elasticity at mean. We then define a differential effect between regular price and discount for brand sales i in category j in store format k as $\hat{\eta}_{1ijk} + \hat{\eta}_{2ijk}$ [4]. We obtain standard errors for estimated regular price elasticity and estimated discount elasticity using the delta method (Greene 2003). For differential effect between regular price and discount, we obtain $SE(\hat{\eta}_{1ijk} + \hat{\eta}_{2ijk}) = \sqrt{Var(\hat{\eta}_{1ijk}) + Var(\hat{\eta}_{2ijk}) + 2Cov(\hat{\eta}_{1ijk}, \hat{\eta}_{2ijk})}$.

The differential effect is calculated to understand the relative impact of regular prices and discounts on sales (at mean) within different store formats and product categories. For instance, if the sum of the elasticities approaches zero, it would indicate that the impact of a regular price increase can be attenuated by an equivalent discount, suggesting a symmetric sensitivity to both pricing strategies. Conversely, a significant deviation from zero in this differential effect could highlight an asymmetry in sales responses to price increases versus discounts, which may vary by store format and category. A positive differential effect indicates that discounts more effectively boost sales than regular price increases diminish them, suggesting discounts have a stronger influence. On the other hand, a negative differential effect implies that regular price increases suppress sales more significantly than discounts can enhance them, reflecting heightened price sensitivity. This hypothesis is based on the assumption that the economic impact of a price increase equals an equivalent discount, thereby leading to a symmetrical effect on sales volume across different store formats and categories. To test this assumption, we conduct a linear hypothesis test to determine if the sum of the estimated coefficient of regular price ($\hat{\beta}_{1ijk}$) and discount ($\hat{\beta}_{2ijk}$) is statistically equal to zero.

5.2 Second-Stage Regression

In the second-stage analysis, we estimate the following regressions for differential effect between regular price and discount ($\hat{\eta}_{1ijk} + \hat{\eta}_{2ijk}$) and discount elasticity ($\hat{\eta}_{2ijk}$) for brand i in category j

[4] Given our research framework, which posits that store formats may influence the relative effectiveness of regular prices and discounts, we hypothesize that the effects of price increases (regular price elasticity) and price reductions (discount elasticity) should ideally offset one another, resulting in a combined sum of these elasticities approaching zero.

offered in store format k:

$$(2) \quad \hat{\eta}_{1ijk} + \hat{\eta}_{2ijk} = \sum_{k=1}^{3} \delta_{0k} Format_{ijk}$$
$$+ \sum_{k=1}^{3} \delta_{1k} Format_{ijk} * BDepth_{ijk} + \sum_{k=1}^{3} \delta_{2k} Format_{ijk} * BBreadth_{ijk}$$
$$+ \sum_{k=1}^{3} \delta_{3k} Format_{ijk} * BFreq_{ijk} + \sum_{k=1}^{3} \delta_{4k} Format_{ijk} * PL_{ij}$$
$$+ \sum_{k=1}^{3} \delta_{5k} Format_{ijk} * CDepth_{jk} + \sum_{k=1}^{3} \delta_{6k} Format_{ijk} * CBreadth_{jk}$$
$$+ \sum_{k=1}^{3} \delta_{7k} Format_{ijk} * CComp_{jk} + \sum_{k=1}^{3} \delta_{8k} Format_{ijk} * Impulse_{j}$$
$$+ \sum_{k=1}^{3} \delta_{9k} Format_{ijk} * Storable_{j} + \upsilon_{ijk}$$

$$(3) \quad \hat{\eta}_{2ijk} = \sum_{k=1}^{3} \kappa_{0k} Format_{ijk}$$
$$+ \sum_{k=1}^{3} \kappa_{1k} Format_{ijk} * BDepth_{ijk} + \sum_{k=1}^{3} \kappa_{2k} Format_{ijk} * BBreadth_{ijk}$$
$$+ \sum_{k=1}^{3} \kappa_{3k} Format_{ijk} * BFreq_{ijk} + \sum_{k=1}^{3} \kappa_{4k} Format_{ijk} * PL_{ij}$$
$$+ \sum_{k=1}^{3} \kappa_{5k} Format_{ijk} * CDepth_{jk} + \sum_{k=1}^{3} \kappa_{6k} Format_{ijk} * CBreadth_{jk}$$
$$+ \sum_{k=1}^{3} \kappa_{7k} Format_{ijk} * CComp_{jk} + \sum_{k=1}^{3} \kappa_{8k} Format_{ijk} * Impulse_{j}$$
$$+ \sum_{k=1}^{3} \kappa_{9k} Format_{ijk} * Storable_{j} + \phi_{ijk}$$

where k is an indication of the store format ($k = 1$ is Hypermarket, $k = 2$ is Supermarket and $k = 3$ is Convenience Store). $\sum_{k=1}^{3} \delta_{0k}$ and $\sum_{k=1}^{3} \kappa_{0k}$ indicate direct effects of store formats on differential effect between regular price and discount and discounts elasticity while $\sum_{n=1}^{9} \sum_{k=1}^{3} \delta_{nk}$ and $\sum_{n=1}^{9} \sum_{k=1}^{3} \kappa_{nk}$ suggest moderating effects of store formats on brand factors and category factors. To account for uncertainty in the elasticity estimate, we estimate model (2) and (3) with weights equal to the inverse standard errors of the elasticity estimates (Datta et al., 2022).

Model (2) examines the combined effect of changes in regular prices and discounts on sales at mean ($\hat{\eta}_{1ijk}+\hat{\eta}_{2ijk}$), aiming to capture how the impact of regular price adjustments and discounting strategies together deviate from zero. The independent variables include store format indicators and their interactions with brand and category factors such as discount depth, discount breadth, discount frequency, private label status, and category competitiveness. The specification allows us to explore how these factors modify the combined impact of pricing strategies across different retail environments. It is worth noting that the potential zeros in differential effects ($\hat{\eta}_{1ijk} + \hat{\eta}_{2ijk}$) do not necessarily signify a lack of data variability or diminish the validity of our models. In our analysis, zeros in the differential effect can indicate instances where the price increase and discount effects are symmetrical, leading to a net zero effect on sales. For example, high variability in the discount depth or frequency among different store formats can help us understand why some formats may see more substantial deviations from zero in the differential effect, indicating more aggressive or effective discounts offered.

Model (3) focuses specifically on the elasticity of discounts ($\hat{\eta}_{2ijk}$), assessing how responsive sales are to changes in discounts within each store format. By incorporating interactions of store formats with brand and category characteristics, these models are specified to identify how different retail environments influence the effectiveness of discounts.

6 Results

6.1 Elasticities Across Store Formats

We report the summary of the first stage analysis (by format) in table 7. In addition to model (1) specified in section 5, we simply run the model with only final price (i.e., $RegPrice + Discount$) as a baseline to compare the estimates and model fits with model (1). When the final price is considered as a single variable, its elasticity appears larger, suggesting a potential overestimation of overall price sensitivity compared to analyzing the separated effects of regular price and discount. Although the error correction models provide relatively good model fits, evidenced by comparable Adjusted R^2 values, it remains inconclusive whether the model that separate the effects of both regular price and discount performs significantly better than the model capturing only the total effect of the final price. The similar Adjusted R^2 values, which account for the number of predictors used, suggest that both models explain a similar proportion of the variance in sales, adjusted for the complexity of each model.

The estimated average of regular price elasticity across all store formats has a significant negative effect, with elasticity values ranging from -4.46 in the hypermarket to -2.74 in the convenience store. The magnitude of this elasticity is more pronounced in the hypermarket (i.e., 1% increase in regular price from mean lead to 4.25% decrease in sales).

The estimated average of discount elasticity shows a significant positive effect across all formats, with the hypermarket exhibiting the most substantial response (9.63), followed by the supermarket (5.87) and the convenience store (1.94). The varying degrees of responsiveness across formats highlight the differential impact of promotional activities, suggesting that while discounts can

Table 7: Summary Statistics for Estimated Price Elasticities and The Difference Between Regular Price and Discount Elasticity of Focal Brand and Category by Store Format

	Hypermarket	**Supermarket**	**Convenience**
N	71	71	38
Final Price Elasticity			
Mean[a]	-7.29***	-5.71***	-2.95***
%Pos	1	1	3
%Neg	80	70	66
%n.s.	18	28	31
Regular Price Elasticity			
Mean[a]	-4.46***	-2.79***	-2.74***
%Pos	2	0	5
%Neg	73	52	58
%n.s.	24	48	37
Discount Elasticity			
Mean[a]	9.63***	5.87***	1.94***
%Pos	75	65	61
%Neg	0	1	0
%n.s.	24	35	39
Differential Elasticity			
Mean[a]	-4.27***	-2.64***	-2.72***
%*s.*[b] for $\hat{\beta}_{1ijk} + \hat{\beta}_{2ijk} = 0$	76	65	61
Model Fit			
R^2 for Final Price Model	0.81	0.77	0.77
R^2 for Reg Price and Discount Model	0.81	0.77	0.78

a: Two-sided tests of significance. Elasticities are weighted by inverse standard errors. Significance tested (at p < .10) using meta-analytic p-value (method of added Zs) (Datta et al., 2022).

b: The linear hypothesis test of equality of regression coefficients ($\hat{\beta}_{1ijk}$ = -$\hat{\beta}_{2ijk}$). Significance tested (at p < .10) using Wald test.

Notes: Reported average of elasticies (standard errors) across 166 brands across 14 categories. The column "%Pos" ("%Neg") shows the percentage of estimated elasticities that are significant and positive (negative) in magnitude. The column "%n.s." reports the share of estimated elasticities that is not significantly different from zero (p ≥ .10). The reported average R^2s are Adjusted R^2s that penalize the addition of parameters. Note that the elasticity estimates for 14 brands have been excluded from the analysis. This exclusion is due to a small number of discounts for these brands, which leads to singularities in the regression model.

boost sales, the extent of their impact can vary depending on the store type.

The differential elasticity, suggesting the combined effect of change in regular prices and discounts, shows significant negative values across all formats, from -4.27 in the hypermarket to -2.72 in the convenience store. The negative differential effects suggest that the impact of regular price increases generally outweighs the positive effects of discounts. Despite the higher average discount elasticity, the combined effect, when considering the relative weights of regular price and discount changes, shows that regular price increases tend to dominate. In other words, the negative effect of regular price increases slightly outweighs the positive effect of discounts in the weighted average calculation. It is worth noting that about half of the brands in this study exhibit a sum of the estimated coefficients of regular price and discount that is statistically not equal to zero. This finding addresses that the effects of price increases and discounts are not symmetric requiring the separate analysis between regular prices and discounts to fully understand their distinct impacts on sales.

The histograms of estimated elasticities for regular price and discount are also presented in figure 6 and 7 respectively. The presented estimates are standardized estimates which take into account the standard error of the estimates. The histograms of estimated price elasticities provide insights into the differential impacts of price changes on brand sales across different store formats. The empirical estimates indicates the shapes of these distributions to be asymmetric across formats. Regular price elasticity predominantly exhibits negative values, with hypermarket showing a particularly strong negative response, indicating a significant sensitivity to price increases. Discount elasticity shows positive values, suggesting a robust responsiveness to promotional pricing, with the hypermarket again displaying the most pronounced positive responses. Both convenience store and supermarket exhibit distributions that are more centered around zero, suggesting a more moderate response to price changes and discounts compared to the hypermarket.

6.2 Differential Effects between Regular Prices and Discounts

We examine how store formats influence brand and category factors in determining brand promotional price effectiveness (i.e., the difference between regular price elasticity and discount elasticity) and discount effectiveness (discount elasticity). We report the results of the second-stage analysis in Table 8.

While there is no main effect of store formats on discount elasticity, significant effects on the differential effects between regular price and discount are found in the convenience store (which is the intercept) and supermarket. The large magnitude of negative intercept implies that the effects of changes in regular prices are significantly more pronounced than the same amount of changes in discounts across all formats (since $\hat{\beta}_{1ijk} + \hat{\beta}_{2ijk} < 0$). The positive coefficient for the supermarket suggests that the differential effects in the supermarket are not much different compared to the convenience store except for the category that usually offers higher discounts. Private labels tend

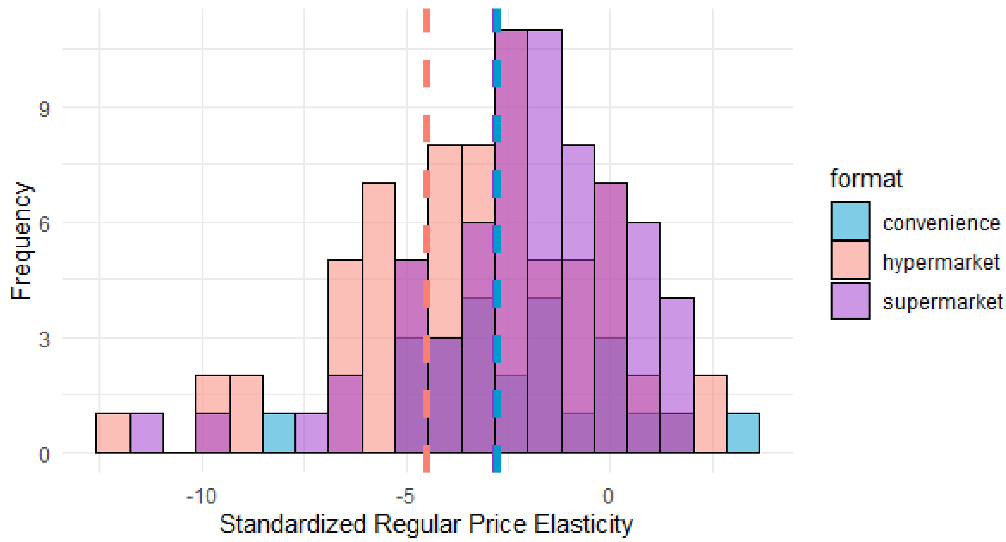

Figure 6: Histogram of Standardized Regular Price Elasticity with Weighted Average by Format (Vertical Line)

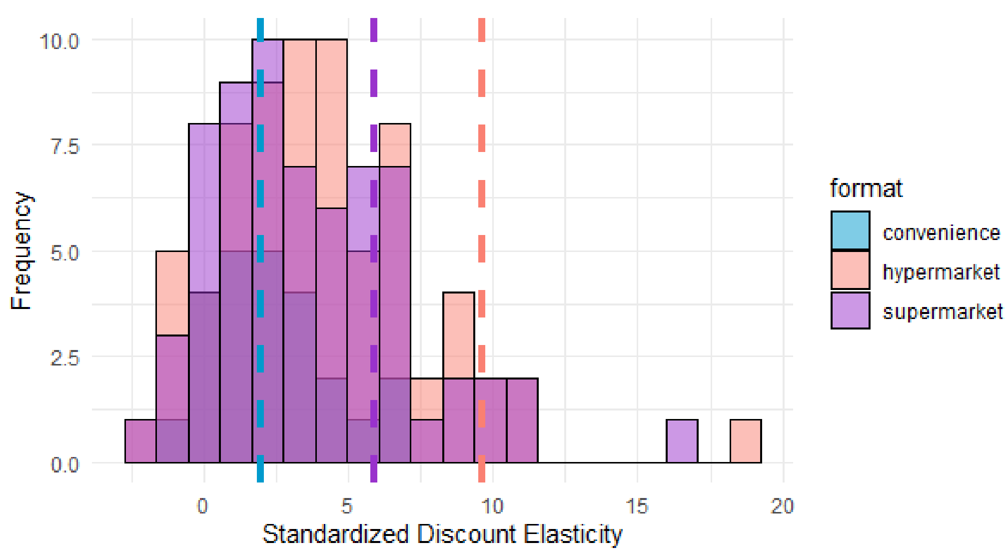

Figure 7: Histogram of Standardized Discount Elasticity with Weighted Average by Format (Vertical Line)

to be more sensitive to discounts, especially in both hypermarket and convenience store (since $\hat{\beta}_{1ijk} + \hat{\beta}_{2ijk} > 0$).

There are no significant direct effects from store format on determining brand discount effectiveness. The store formats tend to moderate few brand factors and category factors including brand discount depth, brand discount breadth, category discount depth and category discount breadth. A deeper discount depth of the brand and category over time in the supermarket can increase an effect of discounts offered. Moreover, while discounts seem to be more effective for brands in categories that generally offer a number of discounts in the hypermarket, the effectiveness are attenuated if these categories usually offer deep discounts. However, there is no evidence of convenience store influencing any effects from these factors.

No moderating effects are observed on brand discount frequency, brand discount variety, competition within the brand's category, or static characteristics of categories which are impulsiveness and storability. Thus, store formats, particularly for the hypermarket and the supermarket, influence the discount effectiveness by the extent of providing appropriate discount activities at the brand level in supermarket and category level in hypermarket. These results address the importance of discount depth and breadth, rather than merely the frequency or variety of offers, in shaping consumer response and enhancing sales effectiveness within these larger store formats.

Table 8: (Second-Stage) Regression of Differential Elasticity and Discount Elasticity on Brand Factors and Category Factors across Store Formats.

Factor	Moderator **Formats**	DV: $\hat{\beta}_{1ijk} + \hat{\beta}_{2ijk}$ Differential Elasticity	DV: $\hat{\beta}_{2ijk}$ Discount Elasticity
Main Effect	(Intercept)	-5362 (2.198)**	0.003 (0.030)
	Hypermarket	1.651 (2.453)	-0.046 (0.040)
	Supermarket	4.659 (2.704)**	-0.049 (0.035)
BDepth	Hypermarket	-0.039 (0.278)	-0.001 (0.005)
	Supermarket	-0.071 (0.358)	0.020 (0.004)***
	Convenience	0.185 (0.463)	0.004 (0.009)
BBreadth	Hypermarket	-0.089 (0.109)	-0.001 (0.001)
	Supermarket	0.016 (0.089)	0.004 (0.002)**
	Convenience	-0.038 (0.136)	-0.003 (0.004)
BFreq	Hypermarket	-0.019 (0.033)	-0.0004 (0.001)
	Supermarket	0.029 (0.039)	-0.0004 (0.001)
	Convenience	-0.022 (0.055)	0.0004 (0.001)
PL	Hypermarket	1.161 (0.431)***	0.019 (0.008)**
	Supermarket	0.627 (0.613)	0.009 (0.007)
	Convenience	2.025 (0.830)**	-0.013 (0.017)
CDepth	Hypermarket	-0.036 (0.257)	-0.014 (0.004)***
	Supermarket	-0.336 (0.315)*	-0.006 (0.004)
	Convenience	-0.035 (0.658)	-0.002 (0.011)
CBreadth	Hypermarket	0.064 (0.064)	0.005 (0.002)***
	Supermarket	-0.017 (0.083)	0.002 (0.001)
	Convenience	0.202 (0.201)	0.0002 (0.003)
CComp	Hypermarket	0.084 (0.099)	0.001 (0.001)
	Supermarket	-0.016 (0.106)	0.003 (0.002)
	Convenience	0.172 (0.116)	-0.001 (0.002)
Impulse	Hypermarket	0.286 (0.462)	-0.003 (0.007)
	Supermarket	-0.175 (0.575)	0.004 (0.007)
	Convenience	0.262 (1.252)	0.012 (0.020)
Storable	Hypermarket	0.507 (0.461)	-0.002 (0.007)
	Supermarket	0.166 (0.502)	0.010 (0.006)
	Convenience	0.416 (0.764)	0.013 (0.013)
Observations		161	161
R-Squared		0.2901	0.6132
Note:			*p<0.1; **p<0.05; ***p<0.01

7 Discussion

Our study investigate the differentiated impacts of regular prices and discounts on brand performance across various categories and store formats within the same retail chain. We specifically addressed two research questions: 1) How do store formats directly influence the differential effectiveness of regular prices and discounts and moderate the impact of brand factors and category factors on these differential effects? and 2) How do store formats directly influence discounts' effectiveness and moderate the impact of brand factors and category factors on the effectiveness of discounts?

To answer these questions, we analyze the performance of 180 brands across 14 frequently purchased categories, which engaged in adjustments of regular prices or the offering of temporary discounts. We obtain brand sales data across format through the loyalty membership database of a Nordic grocery retailer operating under a franchise model with individual store owners. In the following subsections, we discuss our results with respect to the proposed research questions and the implications. In the following subsections, we discuss our results with respect to the proposed research questions as well as broader implications and limitations of our findings.

7.1 Role of Store Formats

The results of our study contribute to the understanding of how different store formats influence the effectiveness of promotional pricing strategies within a multi-format retail setting. Our findings particularly focus on the role of store formats in directly influencing and moderating the impact of brand factors and category factors on the differential effectiveness between regular prices and discounts and the discounts effectiveness.

Firstly, the lack of significant main effects from store formats on discount elasticity contrasts with some prior research (e.g., Haans & Gijsbrechts (2011)), suggesting that discount sensitivity may not be as varied across formats as expected. Haans & Gijsbrechts (2011) highlight that promotional effectiveness tends to decrease as store size increases, particularly noting that larger stores experience diminished sales lifts from discounts compared to smaller stores. This variance could be due to our study's focus on different store formats within the same retail chain, which might not exhibit as pronounced differences in size as those examined by Haans & Gijsbrechts.

However, the findings of store format in affecting discount effectiveness and the differential effects together add complexity to our understanding of consumer behavior. While the type or characteristics of store format may not directly affect how consumers react to discounts, it does influence the differential effects between regular price changes and discounts. This is particularly evident in supermarkets, where we observed a significant differential effect between regular prices and discounts. This effect, which was more pronounced than in other formats, suggests that consumers in supermarkets are responsive to changes in regular prices over discounts. This responsiveness might be enhanced by the smaller size and more curated offerings of supermarkets compared to larger formats like hypermarkets, where the extensive product assortment could dilute the impact of any single discount. This finding may counter some existing assumptions that discounts are universally compelling across all formats. This aligns with our initial discussion in the literature review where we posited that consumer responses to pricing strategies could vary significantly depending on the physical retail environment, echoing theories of store environments. However, it aligns with literature suggesting that different retail environments cater to varying consumer needs, which in turn affects pricing strategy effectiveness (e.g., Jindal et al., 2020).

Our findings indicate a significant sensitivity to temporary discounts changes for private labels within the hypermarket. This effectiveness can be attributed to the extensive product assortment and aggressive pricing strategies that are characteristic of store brands in these large retail environments. In hypermarkets, where consumers are presented with a wide range of options, discounts on private labels can significantly influence purchasing decisions. This suggests that shoppers in hypermarkets are especially responsive to discounts, perhaps because these price reductions make private label products even more attractive compared to their branded counterparts.

We also discovered that store formats moderate certain brand and category factors, including brand discount depth, brand discount breadth, category discount depth, and category discount breadth, with significant implications in hypermarkets and supermarkets. This moderating role suggests that deeper discounts can amplify the effectiveness of promotional strategies within these larger store formats, a finding that supports the literature emphasizing the importance of tailored promotional strategies in different categories in different store environments (e.g., Gauri et al., 2017; Breugelmans et al., 2023). No moderating effects were observed on factors like brand discount frequency or the variety of discounts, which points to a potential saturation in how frequently discounts can influence consumer purchasing behavior across formats. This aspect of the findings suggests that beyond a certain frequency, discounts may cease to have a differential impact, aligning with the notion that too frequent promotions could lead to 'promotion fatigue' among consumers (Srinivasan et al., 2004).

In summary, store formats directly influence the differential effectiveness of regular prices and discounts by making regular price more effective in supermarket. However, only changes of regular prices for private labels offered in hypermarket tend to be more effective compared to the offering temporary discounts. In term of discount effectiveness, store formats do not directly influence the discounts effectiveness but moderate the effects through brand and category factors. There are different moderating effect on brand and categories discount activities.

Grocery retailers can derive practical insights from our findings on the differential impacts of promotional strategies across various store formats. Given the significantly higher sensitivity to regular price adjustments, particularly in supermarkets, retailers should consider emphasizing regular pricing strategies over aggressive discounting for these products, except for private labels in hypermarkets. This approach could prevent consumers from becoming overly conditioned to expect discounts, which might undermine their perception of value and diminish brand loyalty. For supermarkets, where deeper discounts have been shown to enhance promotional effectiveness significantly, retailers could consider designing discount strategies that account for these format-specific responses. Rather than suggesting a shift away from centrally coordinated promotional strategies, our findings highlight that such centralized decisions can benefit from incorporating format-specific differences in consumer responsiveness. However, decisions regarding brand-level discounts should also be aligned with category-level discount activities. A deep brand discount might not be as effective in categories that already exhibit high levels of discounting. The strategy should also ensure that discounting does not become overly frequent to avoid promotion fatigue, which can erode the perceived value of deals over time. Yet, increases in sales volume driven by either reductions in regular prices or discounts do not necessarily translate into increased profitability. Thus, careful consideration of margin impacts and long-term brand value is essential for sustainable financial success.

7.2 Limitations

This study has limitations that suggest opportunities for future research. Firstly, the results are inherently bound by the scope of the data used. Particularly, the grocery retailer data used in our studies predominantly represents loyal customers of a hypermarket, which may not fully capture the potential different shopping behaviors and preferences found in the general population. This suggests caution when generalizing these findings to broader retail contexts or different customer segments. Another significant limitation is the absence of data on additional marketing mix instruments, such as advertising, which are not included in the analysis. This exclusion is particularly relevant as these factors can significantly moderate and influence consumer behavior and purchasing decisions. The lack of this data means that the estimated effects on consumer behavior and sales performance, particularly in the effect of prices might be overestimated (Bijmolt et al., 2005) or not fully representative of the actual market dynamics. Future studies could benefit from incorporating all relevant marketing-mix information and data from a wider range of retail formats, including nontraditional physical stores such as showrooms with different style and installations (Breugelmans et al., 2023). This would enable a more broader and deeper view of consumer behavior across different retail environments and potentially reveal insights into emerging trends and preferences. Another area for future exploration is the operationalization of trip types used as a control variable in our analysis. The classifications of trip types are algorithmically generated based on transaction data, and thus, they serve primarily as proxies for understanding potential shopping patterns and behaviors. This methodological choice allows us to approximate the nature of shopping trips without making direct assumptions about the actual intentions of the shoppers. Consequently, while these trip types provide valuable insights into different shopping patterns, they should be considered conceptual tools rather than definitive indicators of shopper intent. The classifications help control for observed potential relation between shopping trip characteristics and store format, and self-selection bias. Further research could enhance the operationalization of trip types by incorporating direct consumer feedback regarding their shopping intentions, which would allow for a more precise definition of trip types and potentially yield more robust conclusion into how shopping motivations influence responses to pricing strategies and interact across various store formats.

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